

Annotation Transfer Tool

<https://github.com/aiyshwary/Annotation-Transfer-Tool>

Python

Computer Vision

AWS S3

CVAT

PyTorch

OVERVIEW

A smart auto-annotation pipeline that leverages deep image similarity to transfer labels from previously human-annotated images to new unannotated ones, significantly reducing manual labeling effort in computer vision workflows.

IMPACT

Reduced manual annotation effort by automatically transferring labels from existing human-labeled images to new images using deep feature similarity and a modular 8-step pipeline integrated with CVAT and AWS S3.

TECHNICAL WALKTHROUGH

High-level overview

This project is an Annotation Transfer Tool that significantly reduces manual labeling effort by automatically transferring annotations from previously human-labeled images to new images using deep image similarity.

In short

- Find similar images → reuse their annotations → verify visually →
- export back to CVAT
- What problem does it solve?
- Manual annotation is slow and expensive. Even auto-annotation models struggle when data is limited or domain-specific. This tool leverages already annotated images and transfers their labels to new images by identifying visually similar samples using deep learning.
- Core idea
- Instead of predicting bounding boxes from scratch, the tool searches for visually similar images using deep feature embeddings and transfers annotations from the closest matches.

This makes it

- Faster than manual labeling
- More reliable than blind auto-annotation

- Especially effective in low-variation industrial datasets

- End-to-end pipeline explanation

Configuration-driven setup

The entire pipeline is controlled through a single `config.yml`, which defines input paths, S3 locations, and processing parameters.

This makes the pipeline

- Reproducible

- Easy to reconfigure across datasets

Image & annotation ingestion

Images are downloaded from S3 based on a Datumaro JSON file exported from CVAT. Only relevant images are pulled, avoiding unnecessary data transfer.

Scripts involved

- `s3_download.py`

- `datuamaru_crop_sort.py`

Label normalization

Annotations are normalized into a consistent format so that downstream processes like cropping and similarity matching work reliably.

Script

- `normalize_labels.py`

Crop, mask, and label generation

For each annotated object, the tool generates image crops, corresponding label files, and binary masks. This creates a reference library of labeled object appearances.

Script

- `crops_generation.py`

Deep image similarity search

A deep learning-based feature extractor generates embeddings for image crops. Using `DeepImageSearch`, the tool identifies visually similar images across the dataset.

Script

- `CheckSimilarityScore_gpu.py`

This is where

- Feature-based search replaces brute-force labeling
- Previously labeled examples guide new annotations

Annotation transfer & correction

Annotations from the most similar matches are transferred to the target images. Any mismatches or incorrect labels are corrected and mapped back to the original annotation files.

Script

- `Correcting_matching.py`

This step ensures

- Label consistency
- Reduced noise from similarity errors

Visualization for validation

Both normalized and denormalized annotations are visualized to verify correctness before final export.

Scripts

- `visualization_normalized.py`
- `visualization_denormalized.py`
- This acts as a human-in-the-loop sanity check.

Export back to CVAT

Finally, the updated annotations are converted back into Datumaro JSON format, making them directly uploadable to CVAT.

Script

- `fetch_json_data.py`
- Two ways to run the system
- Full automation
- The entire pipeline can be run with a single command (`run.py`) for batch processing.

- Step-by-step execution

- Each stage can also be executed independently for debugging, validation, or partial runs.

- Interactive demo

- A Jupyter notebook (DEMO.ipynb) demonstrates the entire workflow interactively, making it easy to understand intermediate outputs and debug results.

- One-liner summary

- I built a deep similarity-based annotation transfer system that automatically reuses existing human annotations to label new images, reducing manual labeling effort while maintaining annotation quality.

KEY DETAILS

- Downloads labeled images from AWS S3 and extracts annotations from Datamano JSON format.

- Generates image crops, normalizes labels, and creates masks for each labeled region.

- Uses DeepImageSearch (GPU-accelerated) to identify the most visually similar labeled images for each new image.

- Transfers and corrects annotations from matched images, then exports back to CVAT-compatible JSON.

- Includes normalized and denormalized visualization scripts and an interactive Jupyter demo notebook.